

Def. Let  $V$  be a finite-dimensional inner product space over  $F = \mathbb{C}$  or  $\mathbb{R}$ .

A linear operator  $T: V \rightarrow V$  satisfying

$$\forall x \in V, \|T(x)\| = \|x\|$$

is called unitary when  $F = \mathbb{C}$ , orthogonal when  $F = \mathbb{R}$ .

Lemma. Let  $V$  be an inner product space,  $H: V \rightarrow V$  be a Hermitian.

If for all  $x \in V$ ,  $\langle x, H(x) \rangle = 0$ , then  $H = 0$ .

Proof. Let  $x \in V$  be given. Then,

$$\begin{aligned} 0 &= \langle x + H(x), H(x + H(x)) \rangle = \langle x + H(x), H(x) + H^2(x) \rangle \\ &= \langle x, H(x) \rangle + \langle x, H^2(x) \rangle + \langle H(x), H(x) \rangle + \langle H(x), H^2(x) \rangle \\ &= 0 + \|H(x)\|^2 + 0 \end{aligned}$$

Therefore,  $H(x) = 0$ .

Thm. Let  $V$  be a fin. dim. inn. V.s over  $F = \mathbb{C}$  or  $\mathbb{R}$ .

For any linear operator  $T: V \rightarrow V$ ,

$$1) \text{ For all } x \in V, \|T(x)\| = \|x\|$$

$$2) \text{ For any } x, y \in V, \langle T(x), T(y) \rangle = \langle x, y \rangle$$

$$3) T \circ T^* = I$$

$$4) T^* \circ T = I$$

Proof. Suppose 1). we follow along  $1) \Rightarrow 3) \Rightarrow 4) \Rightarrow 1)$ .

Let  $x \in V$  be given. Then

$$\langle x, x \rangle = \langle T(x), T(x) \rangle = \langle x, (T^* \circ T)(x) \rangle \text{ which implies } \langle x, (I - T^* \circ T)(x) \rangle = 0.$$

And, since  $(I - T \circ T^*)^* = I^* - (T \circ T^*)^* = I - T^* \circ T$ , it is Hermitian, therefore,  $I = T^* \circ T$ , by the Lemma. Clearly, this implies  $I = T \circ T^*$ .

Finally, if  $T^* \circ T = I$ , then for any  $x \in V$ ,

$$\langle x, x \rangle = \langle (T^* \circ T)(x), x \rangle = \langle T(x), T(x) \rangle$$

$$I_n \quad F = \mathbb{C}.$$

